

Evidence

1 - Terraform plan output

```
my_ubuntu_docker - sergio@sergio: ~/terraform-assessment - Xshell 8 (Free for Home/School)
File Edit View Tools Tab Window Help
ssh://sergio:*****@192.168.90.166:22
To add the current session, click on the left arrow button.
1 my_ubuntu_docker x +
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
sergio@sergio:~/terraform-assessment$ terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are
indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_eip.bastion_eip will be created
+ resource "aws_eip" "bastion_eip" {
  + allocation_id      = (known after apply)
  + arn                = (known after apply)
  + association_id     = (known after apply)
  + carrier_ip         = (known after apply)
  + customer_owned_ip  = (known after apply)
  + domain             = "vpc"
  + id                 = (known after apply)
  + instance           = (known after apply)
  + ipam_pool_id       = (known after apply)
  + network_border_group = (known after apply)
  + network_interface  = (known after apply)
  + private_dns        = (known after apply)
  + private_ip         = (known after apply)
  + ptr_record         = (known after apply)
  + public_dns         = (known after apply)
  + public_ip          = (known after apply)
  + public_ipv4_pool    = (known after apply)
  + region             = "us-east-1"
  + tags_all           = (known after apply)
}

# aws_eip.nat_1 will be created
+ resource "aws_eip" "nat_1" {
  + allocation_id      = (known after apply)
  + arn                = (known after apply)
  + association_id     = (known after apply)
  + carrier_ip         = (known after apply)
  + customer_owned_ip  = (known after apply)
}
```

To add the current session, click on the left arrow button.

1 my_ubuntu_docker x +

```
# aws_vpc_security_group_ingress_rule.web_http_from_alb will be created
+ resource "aws_vpc_security_group_ingress_rule" "web_http_from_alb" {
+   arn                        = (known after apply)
+   from_port                 = 80
+   id                        = (known after apply)
+   ip_protocol               = "tcp"
+   referenced_security_group_id = (known after apply)
+   region                    = "us-east-1"
+   security_group_id         = (known after apply)
+   security_group_rule_id     = (known after apply)
+   tags_all                  = {}
+   to_port                   = 80
+ }

# aws_vpc_security_group_ingress_rule.web_ssh_from_bastion will be created
+ resource "aws_vpc_security_group_ingress_rule" "web_ssh_from_bastion" {
+   arn                        = (known after apply)
+   from_port                 = 22
+   id                        = (known after apply)
+   ip_protocol               = "tcp"
+   referenced_security_group_id = (known after apply)
+   region                    = "us-east-1"
+   security_group_id         = (known after apply)
+   security_group_rule_id     = (known after apply)
+   tags_all                  = {}
+   to_port                   = 22
+ }
```

Plan: 43 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ load_balancer_dns_name = (known after apply)
+ vpc_id                 = (known after apply)
```

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

sergio@sergio:~/terraform-assessment\$

2 - Terraform apply completion

```
my_ubuntu_docker - sergio@sergio: ~/terraform-assessment - Xshell 8 (Free for Home/School)

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ssh://sergio:*****@192.168.90.166:22

To add the current session, click on the left arrow button.

1 my_ubuntu_docker x +

sergio@sergio:~/terraform-assessment$ terraform apply -auto-approve

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_eip.bastion_eip will be created
+ resource "aws_eip" "bastion_eip" {
+   allocation_id      = (known after apply)
+   arn                 = (known after apply)
+   association_id      = (known after apply)
+   carrier_ip          = (known after apply)
+   customer_owned_ip   = (known after apply)
+   domain              = "vpc"
+   id                  = (known after apply)
+   instance            = (known after apply)
+   ipam_pool_id        = (known after apply)
+   network_border_group = (known after apply)
+   network_interface    = (known after apply)
+   private_dns         = (known after apply)
+   private_ip          = (known after apply)
+   ptr_record          = (known after apply)
+   public_dns          = (known after apply)
+   public_ip           = (known after apply)
+   public_ipv4_pool     = (known after apply)
+   region              = "us-east-1"
+   tags_all            = (known after apply)
+ }

# aws_eip.nat_1 will be created
+ resource "aws_eip" "nat_1" {
+   allocation_id      = (known after apply)
+   arn                 = (known after apply)
+   association_id      = (known after apply)
+   carrier_ip          = (known after apply)
+   customer_owned_ip   = (known after apply)
+   domain              = "vpc"
+   id                  = (known after apply)
+   instance            = (known after apply)
+ }

ssh://sergio@192.168.90.166:22

my_ubuntu_docker - sergio@sergio: ~/terraform-assessment - Xshell 8 (Free for Home/School)

File Edit View Tools Tab Window Help

ssh://sergio:*****@192.168.90.166:22

To add the current session, click on the left arrow button.

1 my_ubuntu_docker x +

aws_nat_gateway.nat_1: Still creating... [00m50s elapsed]
aws_nat_gateway.nat_2: Still creating... [00m50s elapsed]
aws_lb_app_alb: Still creating... [00m50s elapsed]
aws_nat_gateway.nat_1: Still creating... [01m00s elapsed]
aws_lb_app_alb: Still creating... [01m00s elapsed]
aws_nat_gateway.nat_2: Still creating... [01m00s elapsed]
aws_nat_gateway.nat_1: Still creating... [01m10s elapsed]
aws_nat_gateway.nat_2: Still creating... [01m10s elapsed]
aws_lb_app_alb: Still creating... [01m10s elapsed]
aws_nat_gateway.nat_1: Still creating... [01m20s elapsed]
aws_lb_app_alb: Still creating... [01m20s elapsed]
aws_nat_gateway.nat_2: Still creating... [01m20s elapsed]
aws_nat_gateway.nat_1: Still creating... [01m30s elapsed]
aws_lb_app_alb: Still creating... [01m30s elapsed]
aws_nat_gateway.nat_2: Still creating... [01m30s elapsed]
aws_nat_gateway.nat_2: Creation complete after 1m32s [id=nat-04be4aab6484bd077]
aws_route_table.private_rt_2: Creating...
aws_route_table.private_rt_2: Creation complete after 3s [id=rtb-09157e30d4d7d0574]
aws_route_table_association.private_2: Creating...
aws_route_table_association.private_2: Creation complete after 1s [id=rtbassoc-00e7137dedad154c7]
aws_nat_gateway.nat_1: Still creating... [01m40s elapsed]
aws_lb_app_alb: Still creating... [01m40s elapsed]
aws_nat_gateway.nat_1: Creation complete after 1m43s [id=nat-0e7916e9064263820]
aws_route_table.private_rt_1: Creating...
aws_route_table.private_rt_1: Creation complete after 2s [id=rtb-0665771a7091dfa2f]
aws_route_table_association.private_1: Creating...
aws_route_table_association.private_1: Creation complete after 1s [id=rtbassoc-0604e857448bac41f]
aws_lb_app_alb: Still creating... [01m50s elapsed]
aws_lb_app_alb: Still creating... [02m00s elapsed]
aws_lb_app_alb: Still creating... [02m10s elapsed]
aws_lb_app_alb: Still creating... [02m20s elapsed]
aws_lb_app_alb: Creation complete after 2m28s [id=arn:aws:elasticloadbalancing:us-east-1:884057991176:loadbalancer/app/techcorp-alb/0a6b2efc5ab2ec84]
aws_lb_listener.http: Creating...
aws_lb_listener.http: Creation complete after 1s [id=arn:aws:elasticloadbalancing:us-east-1:884057991176:listener/app/techcorp-alb/0a6b2efc5ab2ec84/f62137b83e8bd1c8]

Apply complete! Resources: 43 added, 0 changed, 0 destroyed.

Outputs:

load_balancer_dns_name = "techcorp-alb-1147029791.us-east-1.elb.amazonaws.com"

ssh://sergio@192.168.90.166:22
```

3 - AWS Console showing created resources

Instances EC2

Instances (4) Info

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	web-2	i-009932cbfd4f47a4	Running	t3.micro	3/3 checks passed	us-east-1b	-
☐	db	i-07cfbaf28ea63b37b	Running	t3.small	3/3 checks passed	us-east-1a	-
☐	web-1	i-0b61fc48970787300	Running	t3.micro	3/3 checks passed	us-east-1a	-
☐	bastion	i-0b008a7c6baea5012	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-3-227-140-30.com...

Load Balance

Load balancers (1) What's new?

Actions

Create load balancer

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones
☐	techcorp-alb	Active	application	Internet-facing	IPv4	vpc-026be86857c5d4920	2 Availability Zones

Security Groups

Security Groups (6) Info

Actions

Export security groups to CSV

Create security group

Find security groups by attribute or tag

	Name	Security group ID	Security group name	VPC ID	Description
☐	-	sg-0c77626bd04211298	web-sg	vpc-026be86857c5d4920	Managed by Terrafo
☐	-	sg-0adfa71c4b833f184	db-sg	vpc-026be86857c5d4920	Managed by Terrafo
☐	-	sg-022960badc4860fb7	default	vpc-03e772fb65d336cb4	default VPC security
☐	-	sg-0e287a33427f18ad1	bastion-sg	vpc-026be86857c5d4920	Managed by Terrafo
☐	-	sg-0310715dad4471173	alb-sg	vpc-026be86857c5d4920	Managed by Terrafo
☐	-	sg-00cde7ed4628915c5	default	vpc-026be86857c5d4920	default VPC security

Select a security group

Target Group

Target groups (1/1) Info | What's new?

Actions

Create target group

Filter target groups

	Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
☒	web-tg	arn:aws:elasticloadbalancin...	80	HTTP	Instance	techcorp-alb	vpc-00d9a...

Elastic IP

Elastic IP addresses (1/1) Info

Actions

Allocate Elastic IP address

Find elastic IP addresses by attribute or tag

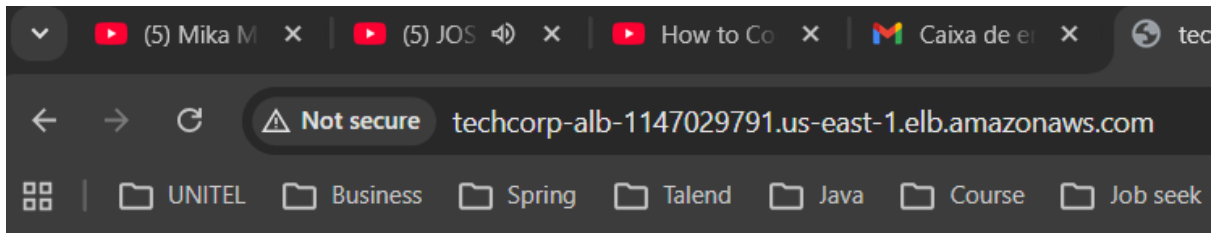
Associated instance ID = i-0dbac49519b3ed1af

Clear filters

	Allocated IPv4 addr...	Type	Allocation ID	Reverse DNS record	Associated instance ID
	18.232.225.209	Public IP	eipalloc-0ea8ec4f163a18dde	-	i-0dbac49519b3ed1af

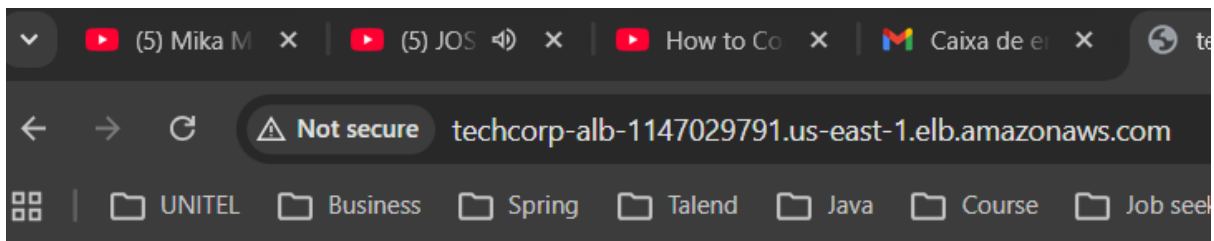
4 - Load balancer serving web pages from both instances

server-1



Web Server OK - ip-10-0-3-171.ec2.internal

server-2



Web Server OK - ip-10-0-4-171.ec2.internal

5 - SSH access through bastion host

```
ec2-user@ip-10-0-1-103:~  
Microsoft Windows [Version 10.0.26100.8457]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\sergi\Downloads>ssh -i techcorp-key.pem ec2-user@18.232.225.209  
Last login: Wed May 20 14:08:47 2026 from 169.239.12.38  
  
  _ _ | _ _ | _ )  
 _ | ( / Amazon Linux 2 AMI  
---| \---|---|  
  
https://aws.amazon.com/amazon-linux-2/  
51 package(s) needed for security, out of 96 available  
Run "sudo yum update" to apply all updates.  
[ec2-user@ip-10-0-1-103 ~]$ |
```

Web-1

```

ec2-user@ip-10-0-3-171:~
Microsoft Windows [Version 10.0.26100.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sergi\Downloads>ssh -i techcorp-key.pem ec2-user@18.232.225.209
Last login: Wed May 20 14:08:47 2026 from 169.239.12.38

  __|  __|_ )
  _| (    /   Amazon Linux 2 AMI
 ---|\---|---|

https://aws.amazon.com/amazon-linux-2/
51 package(s) needed for security, out of 96 available
Run "sudo yum update" to apply all updates.
[ec2-user@ip-10-0-1-103 ~]$ ssh -i key.pem ec2-user@10.0.3.171
Warning: Identity file key.pem not accessible: No such file or directory.
Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-10-0-1-103 ~]$ ssh -i techcorp-key.pem ec2-user@10.0.3.171
Last login: Wed May 20 14:10:10 2026 from ip-10-0-1-103.ec2.internal

  #_
  ' \_   #####_   Amazon Linux 2
 ~ ~ \_   #####\
 ~ ~ \_   ###|
 ~ ~ \_   #/  ---
 ~ ~   V~'  ' ->
 ~ ~ ~
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 ~ ~ _/_/_/
 ~ ~ _/m/'

  A newer version of Amazon Linux is available!

  Amazon Linux 2023, GA and supported until 2028-03-15.
  https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-10-0-3-171 ~]$

```

web-2

```
[ec2-user@ip-10-0-4-171:~]
#_
~\   #####_
~~ \_#####\
~~  \###|
~~   \|#/
~~   V~' '-->
      ~~~
      ~.._.
      _/_/_/_/_/
      _/m/'

Amazon Linux 2

AL2 End of Life is 2026-06-30.

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-10-0-3-171 ~]$ exit
logout
Connection to 10.0.3.171 closed.
[ec2-user@ip-10-0-1-103 ~]$ ssh -i techcorp-key.pem ec2-user@10.0.4.171
^C
[ec2-user@ip-10-0-1-103 ~]$ ssh -i techcorp-key.pem ec2-user@10.0.4.171
Last login: Wed May 20 14:13:15 2026 from ip-10-0-1-103.ec2.internal
#_
~\   #####_
~~ \_#####\
~~  \###|
~~   \|#/
~~   V~' '-->
      ~~~
      ~.._.
      _/_/_/_/_/
      _/m/'

Amazon Linux 2

AL2 End of Life is 2026-06-30.

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-10-0-4-171 ~]$
```

7 - Connect to the postgres instance on the DB server

```
ec2-user@ip-10-0-3-70:~  
[ec2-user@ip-10-0-3-70 ~]$ sudo -u postgres psql  
could not change directory to "/home/ec2-user"  
psql (9.2.24)  
Type "help" for help.  
  
postgres=# |
```